

As many novice scientists learn in their introductory electrical engineering courses, one of the best electrical conductors is copper. The first time that I understood the potential of this element was in 2008, when I won the regional science fair using coal and copper to reproduce the water electrolysis and split the oxygen and hydrogen from water. From that moment onwards, copper was present for the rest of my life: in my high school years, when I learned about electricity and automatization, and years later, when I pursued a Bachelor's in Electrical Engineering at the University of Costa Rica. Over the years, many things changed, and I went through some difficult times when my father passed away, resulting in my first experiences with therapy and mental health support. However, my passion for technology, science, and this popular element of copper was always there in my life, reminding me of its ability to connect powerful currents.

Here in Latin America, as in many other regions of the world, mental health is still taboo. People often do not talk about therapy and deny any kind of psychological intervention due to mental health stigma, even if there's extensive evidence of its benefits. The culture of this region plays a role in shaping and enforcing the stigma, driven by two main factors: family and religion. Strong traditional family values often have a negative impact on seeking mental health support, especially when diagnosis could bring "shame on the family". In addition, religious beliefs also shape the stigma. Patients with chronic mental health conditions or their families often interpret the medical condition as a punishment for misbehavior. For this reason, people avoid talking about mental health. Personally, I became aware of the magnitude of this problem through several years of volunteering supporting high school students from some of the most deprived neighborhoods in Costa Rica. There, I witnessed the impact of the mental health stigma when I realized that a large number of students suffer physical and physiological abuse at home, but usually avoid expressing their feelings about the abuse because talking about emotions is generally perceived as a weakness. As a consequence, they often build a strong personality to survive in their families, leaving no room for vulnerability. Extensive research has shown that an effective approach to alleviate this problem is enabling early access to mental-health support.

There is a pressing need to find new ways to support mental health in my region. Thus, my goal is to reduce the barrier caused by stigma, by increasing people's openness to psychological interventions through technology. The idea is to offer people the possibility to create spaces free of judgment, so they can explore and, eventually, change their misconceptions about mental health. Psychoeducational interventions and Cognitive Behavioral Therapy have proven successful to support various mental health conditions, and technology could provide access to them without feeling exposed to uncomfortable public situations. In this way, we will be reducing the barrier to the first contact with help, which eventually may facilitate receiving support from a professional.

My experience working at the Media Lab has inspired me to design new solutions to support people in identifying and avoiding behaviors that could affect their health. This is similar to "Saving Face", a project in which I participated, headed by Dr. Camilo Rojas, and advised by Dr. Pattie Maes, Dr. Fadel Adib, Dr. Kevin Esvelt, and Dr. Joseph A. Paradiso. We developed a system that leverages commercial earphones and smartphones to identify when the user is about to touch their face, and alert them in order to avoid the action. This results in a reduction in the spread of infections like influenza or COVID-19. My second project at the Media Lab, "Project Us" (which I have been co-leading with Dr. Camilo Rojas), seeks to teach people about emotions and how they can have an impact in their life. Recognizing emotions helps us to be in control of them and helps us to improve our reactions to certain situations. "Project Us" is a platform that uses artificial intelligence to share real-time feedback about users' emotions. We have been using it to support individuals in measuring and improving inclusive behaviors.

To fulfill my goal of reducing the mental health stigma in my region by using technology, I want to pursue a degree in Computer Science with a specialization in Human Computer-Interaction (HCI) at Stanford University. I strongly believe that studying a Master's program related to HCI, where I could learn from and with people of different technical, social, and cultural backgrounds, is key for

my future. A graduate degree in CS combined with HCI give me the bases to design devices for use in everyday life. It will also give the knowledge me to design novel technology-psychological interventions to reduce the gap caused by mental health stigma. Diving deep into HCI at Stanford University, I am excited to take courses like Designing for Accessibility, User Experience Design Project, Social Computing, and Human-Computer Interaction: Foundations and Frontiers. I believe that these courses will provide me with the necessary tools to design novel psychological interventions to address the problem with the stigma in my region.

As a researcher I see the Institute for Human-Centered Artificial Intelligence as a space where I could enhance the effectiveness of psychological interventions by increasing the openness of the user. I would like to create systems that leverage widely available commercial platforms (e.g., smartphones or laptops) to perform longitudinal monitoring of users' emotions during their daily routine. Such systems would enhance the accuracy of diagnosis of mental health challenges and deliver just-in-time psychological interventions tailored for each user. The key difficulty to be addressed is to enable consumer devices to seamlessly track in real-time biosignals that contain information about emotions (e.g., heart rate, breathing rate, body movement patterns, etc.). I plan to begin tackling this challenge by implementing an ultrasound sonar system capable of monitoring the breathing and heart rate of a user from their own laptop. The base for this system will be the FMCW + Doppler signal processing pipeline that we developed for Saving Face, which will be enhanced by factoring in the phase information obtained from the FMCW spectrogram, time-multiplexing the transmission of ultrasound via each speaker of the laptop, and leveraging the spectrograms received in each microphone of the laptop. The result will be an improvement of the accuracy in biosignal measurement, plus enhanced robustness against confounding movements. An interesting demonstration of the system will be running a longitudinal test where the target biosignals of a subject will be monitored with the laptop across multiple days and with wearable devices (to provide a ground truth), while the tone of voice and facial expressions are monitored with the laptop. I expect this setup to exhibit correlations between emotional cues and the biosignals obtained through the wireless system. A future step could be studying the integration between ultrasound and RF signals (e.g., WiFi or Bluetooth), as a mechanism to further improve accuracy and robustness.

Finally, this prestigious and well-connected university is the perfect platform for launching a start-up that could reach Latin America upon my return to Costa Rica. I have been part of some local and international programs related to entrepreneurship, and all of them have brought me to the conclusion that entrepreneurship is the solution to a lot of problems in my country. However, it is important to remember that Latin America is a region with strong social inequalities. This means that not all people could have access to these solutions. Social entrepreneurship could help me transform my solution into a product that people could use outside the lab. By designing a business model that could help people while the products continue to improve it is possible to reduce the economic barrier for access. For this reason, I see the connection between industry and research the Stanford offers as an opportunity to launch my own startup. For these reasons, if given the opportunity, I would like to take courses like Startup Garage to compliment my HCI knowledge

I want to be a conductor, much like copper, that connects the power of HCI and technology with the needs of my region, to make tools and design social programs that will make a difference in the current situation of my country. The Master of Science in Computer Science at Stanford University will teach me the foundations and bases to start my ideas. Working with international and multidisciplinary students will also provide me with innovative ways of making people feel more comfortable with psychological interventions. Social entrepreneurship, combined with HCI, is going to make a difference in the future, and the entrepreneurship ecosystem that Stanford offers to their students will be the first step in my journey to create a company later down the line once I return from my graduate studies.